

Properties of Materials

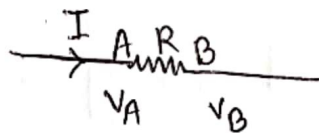
Electrical properties

Ohm's law

At constant temperature, the electricity passing through a circuit is proportional to the potential difference between two ends of the circuit and reciprocal to the resistance of the circuit.

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Let,

A and B are two ends of a circuit and V_A, V_B are the potential difference between two ends. If I ampere electricity is flowing through the circuit then,

$$I \propto (V_A - V_B)$$

$$\Rightarrow I = k(V_A - V_B)$$

$$\Rightarrow I = \frac{1}{R} V \quad \left[\because k = \frac{1}{R} \right]$$

$$\Rightarrow V = IR$$

Electric resistivity

Electric resistivity is a fundamental property of a material that measures how strongly it resists electric current.

* Electrical resistivity depends on resistance, cross-sectional area and distance between measuring points.

$$R \propto \frac{l}{A}$$

$$\Rightarrow R = \frac{\rho l}{A}$$

$$\Rightarrow \rho = \frac{RA}{l}$$

$$\Rightarrow \rho = \frac{VA}{Il}$$

$$\text{Unit} = \Omega \cdot \text{m}$$

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Electrical conductivity

Electrical conductivity is a fundamental property of a material that measures the ability of a material to carry an electrical current.

$$\sigma = \frac{1}{\rho}$$

$$\# P = I^2 R = \frac{I^2 \times \rho l}{A}$$

$$\text{unit: } \Omega^{-1} \text{m}^{-1} / \text{S m}^{-1}$$

$$\Rightarrow P = \frac{I^2 l}{\sigma A} \Rightarrow \sigma = \frac{I^2 l}{PA}$$

Current density

Current density (J) is the amount of electric current flowing per unit cross sectional area of a material.

$$J = \frac{E}{\rho}$$

E = Electric field

$$\Rightarrow J = E\sigma$$

$$\Rightarrow J = \frac{I}{A}$$

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Unit : A/m^2

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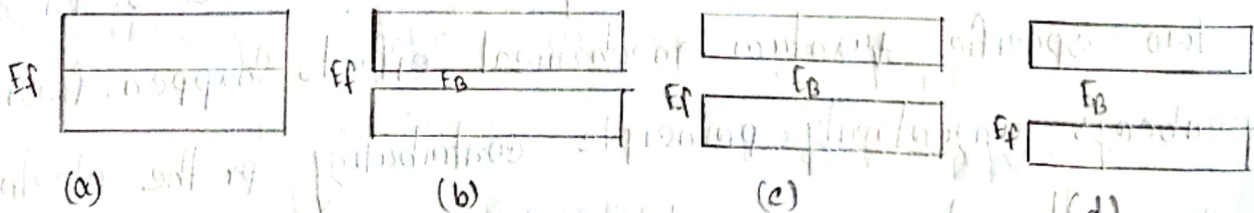
Energy band structures in solid

When atoms come together to form a solid, their valence electrons interact due to forces and they also feel the electronic field produced by their own nucleus. In addition to two specific quantum mechanical effects happen, first by Heisenberg's uncertainty principle contributing to the electrons to a small volume raises their energy and the second effect due to the Pauli exclusion principle limits. As a result, the valence electrons which are delocalized interact and interpenetrate each other. So that their original sharp atomic energy levels are broadened into wider regions called

energy bands. The inner electrons don't form bonds since they are separated from the valence electrons. The bonds are separated by gaps where electron can't exist. The precise location of the band and band gaps depends on the types of atoms.

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The electron energy band structure for the a solid material at equilibrium interatomic separation, the energy band containing the valence electrons is called valence band and the unfilled energy levels into which electrons can be excited to become conductive electrons is called conduction band. The energy corresponding to the highest filled state at 0 K is called fermi energy.



The various possible electron band structures in solids at 0 K. (a) The electron band structure found in metals such as copper, in which there are available electron states above and adjacent to filled states, in the same band.

(b) The electron band structure of metals such as magnesium, in which there is an overlap of filled and empty outer band.

(c) The electron band structure found in the semiconductors, which is the same as the insulators except that the band gap is relatively narrow ($< 2 \text{ eV}$)

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(d) The electronic band structure characteristic of insulators, the filled valence band is separated from the empty conduction band by a relatively large band gap ($> 2 \text{ eV}$)

Thermal properties

Heat capacity

Heat capacity is a fundamental thermal property of a material that ~~assess~~ measures the ability of a material to absorb heat from the external surroundings.

$$C = \frac{dQ}{dT}$$

unit cal/mol K , J/mol K

where dQ is the energy required to produce dT temperature change.

Specific heat capacity

Specific heat capacity is defined as the quantity of heat absorbed per unit mass of the material when its temperature increases 1K . Unit - J/kg K , cal/kg K

* C_p = The heat capacity at constant pressure

C_v = The heat capacity at constant volume.

When heat is added to a gas at constant volume we have,

$$Q_v = C_v dT = \Delta U + W = \Delta U$$

because no work is done. Therefore,

$$dU = C_v dT \Rightarrow C_v = \frac{dU}{dT}$$

When heat is added at constant pressure,

we have,

$$Q_p = C_p dT = \Delta U + W = \Delta U + P \Delta V$$
$$= C_v dT + P dV$$

$$\therefore C_p dT = C_v dT + P dV$$

$$\Rightarrow C_p dT = C_v dT + nR dT \quad [PV = nRT]$$

$$\Rightarrow C_p = C_v + nR$$

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$\therefore$ For an ideal gas, the heat capacity at constant pressure is greater than that at constant

volume by the amount nR .

Thermal expansion

Most solid materials expand upon heating and contract when cooled. This change in length with temperature for a solid may be expressed as follows:

$$\frac{l_f - l_0}{l_0} = \alpha_l (T_f - T_0)$$

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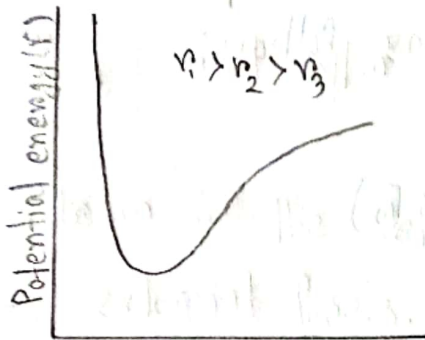
$$\frac{\Delta l}{l_0} = \alpha_l \Delta T$$

where l_0 and l_f represent, respectively, initial and final lengths with the temperature change from T_0 to T_f . The parameter α_l is called the linear co-efficient of thermal expansion. It is a material property that is indicative of the extent to which a material expands upon heating and has units of reciprocal temperature $(^\circ\text{C})^{-1}$ or $(^\circ\text{F})^{-1}$. Heating or cooling affects all the dimension of a body, with a resultant change in volume. Volume changes with temperature may be computed from

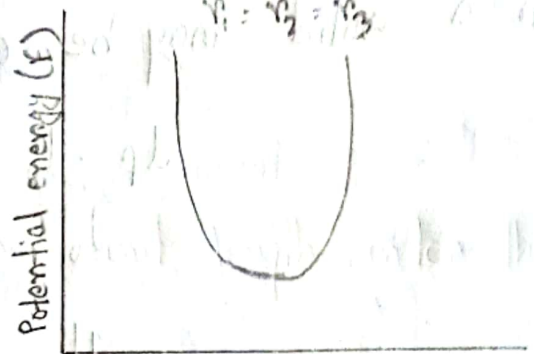
$$\frac{\Delta V}{V_0} = \alpha_v \Delta T$$

where, ΔV and V_0 are the volume change and the original volume, respectively and α_v is the volume coefficient of

thermal expansion. In many materials, the value of α_v is anisotropic i.e it depends on the crystallographic direction along which it is measured. For materials in which the thermal expansion is isotropic, α_v is approximately $3\alpha_l$.



Interatomic distance (r)



Interatomic distance (r)

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Thermal conductivity

Thermal conduction is the phenomenon by which heat is transported from high to low temperature regions of a substance. The property that characterizes the ability of a material to transfer heat is the thermal conductivity.

$$q = -k \frac{dx}{dt} \quad q = -k \frac{dT}{dx}$$

where, q denotes the heat flux or heat flow per unit time per unit area, k is the thermal conductivity and $\frac{dT}{dx}$ is the temperature gradient through the conducting medium.

units of q and k are W/m^2 and W/m K ($\text{Btu/ft}^2\text{-h}^\circ\text{F}$)

The mechanism of heat conduction

Heat is transported in solid materials by both lattice vibrations waves and free electrons. A thermal conductivity is associated with each of these mechanisms and the total conductivity is the sum of the two contributions.

$$k = k_1 + k_2$$

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where, k_1 and k_2 represent the lattice vibration and electron thermal conductivities, respectively, usually one or the other predominates. The thermal energy associated with phonons or lattice waves is transported in the direction of their motion. The k_1 contribution results from a net movement of phonons from high to low temperature regions of a body across which a temperature gradient exists.

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Thermal stress

Thermal stresses are stresses induced in a body as a result of changes in temperature. An understanding of the origins and nature of thermal stresses is important because these stresses can lead to fracture or undesirable plastic deformation.

$$\sigma = E\alpha_l(T_0 - T_f) = E\alpha_l\Delta T$$

α_l = linear coefficient of thermal expansion

E = modulus of elasticity

Mechanical properties

Stress: Stress is the internal resistance set up in a material under the action of external forces.

$$\text{Stress} = \frac{F}{A}$$

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unit N/m^2 or Pa

Strain: Strain is the deformation per unit length under the action of external forces.

$$\text{Strain} = \frac{l_f - l_0}{l_0}$$

* Stress is internal

* Applying load $\longrightarrow$ strain $\longrightarrow$ stress

Tensile stress

The external force per unit area of the material resulting in the stretch of the material is known as tensile stress.

$$\sigma = \frac{F}{A}$$

Tensile strain

Tensile strain is defined as the deformation of a

solid body due to the application of a tensile force or stress.

$$\epsilon = \frac{l_1 - l_0}{l_0}$$

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Compressive stress

Compressive stress is the force that is responsible for the deformation of the material such that the volume of the material reduces.

$$\frac{F}{A} = \text{Compressive stress}$$

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Compressive strain

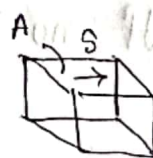
Compressive strain is defined as the deformation in the solid due to the application of compressive strain.

$$\text{Compressive strain} = -\frac{\Delta l}{l_0}$$

Shear stress

Shear stress is the component of stress coplanar with a material cross section.

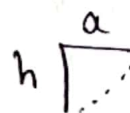
$$\text{Shear stress} = \frac{S}{A} \quad S = \text{Shear force}$$



Shear strain

Shear stress is defined as the ratio of the displacement of a layer to its distance from the fixed layer.

$$\text{Shear strain } \gamma = \frac{a}{h} = \tan \theta$$



Engineering stress-strain

Engineering stress is the applied force divided by the undeformed area over which the force is applied.

Engineering strain is defined as the change in length divided by the original length.

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True stress strain

True stress is ^{the} applied load divided by the actual cross-sectional area (the changing area with respect to time) of the specimen at the load.

$$\sigma_T = \frac{F}{A_i}$$

True strain is the natural logarithm of the ratio of the instantaneous gauge length to the original gauge length.

$$\epsilon_T = \ln \frac{l_i}{l_0}$$

* If no volume change occurs during deformation,

$$l_0 A_0 = l_i A_i$$

$$\Rightarrow \frac{l_0}{l_i} = \frac{A_i}{A_0}$$

Elasticity

A liquid or gas adopts itself to the shape of its container but a solid has a shape of its own which it tends to preserve.

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Loading a solid will change its dimensions but the resulting deformation will disappear upon unloading. The tendency of a deformed solid to seek the original dimensions upon unloading is described to a property called elasticity.

The term 'elasticity' may be defined as the property of a metal by virtue of which it is able to retain its original shape and size after the removal of load. In nature no material is perfectly elastic.

Steel and some other metals have a wide range over which they appear to be elastic.

Plasticity

Plasticity is that property of a material by virtue of which it may be permanently deformed when it has been subjected to an externally applied force great enough to exceed the plastic limit.

The mechanism of plastic deformation is essentially different in crystalline materials and amorphous materials. Crystalline materials undergo plastic deformation as a result of slip along definite crystallographic planes whereas in amorphous materials, plastic deformation occurs when individual molecules or groups of molecules slide past one another.

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